CAUSE & EFFECT MATRIX — LCP-01 Scrubber & Burner Control

Project: Burner PetroBaker | Panel: LCP-01 | Ref. drawing: PH-LCP-01 V1 | Document: PH-LCP-01-CE | Rev. A (for client review) | 26-10-2026

Purpose	Defines the PLC (LCP-01) response to every process and safety cause, as requested in the client e-mail of 04-Oct-2026 (Eng. Amr Farag).
Scope	Gas scrubber alarms/trips, ESD, burner BR01 (Riello RS100) and BR02 (DLX 53 SHFP) Auto/Manual control, tank temperature control (client sheet 'Temperature control of burner'), diesel-mode indication.
Not in scope (Rev. A)	Communication with the Master PLC in the control room — deferred to a later revision. Scrubber low level switch — not installed (DI I12.5 reserved).
Burner flame safety	Purge, ignition, flame supervision and the pilot / main gas valves are handled by each burner's own certified control box (RS100: RMG88; DLX 53: SQ DUAL). LCP-01 only gives the burner its heat request (TL / L) and stage request (TR / T2), and reads its RUN and LOCKOUT signals. LCP-01 never drives a gas valve directly.
Hardwired safety	E-STOP S0 opens the 230 V control supply of BR01 (NC2) and BR02 (NC3) directly, independent of the PLC. Safety thermostat TS (manual reset) is in series with the BR01 control supply. These act even if the PLC fails.
Fail-safe outputs	All PLC outputs de-energise on PLC STOP or 24 VDC loss. Valve types were chosen so that de-energised = safe: XV-401 flare (NO) OPENS, XV-402 drain (NO) OPENS, XV-403 suction (NC) CLOSES, both burners lose their heat request.
How to read the matrix	One row per cause. Cells under each effect say what happens to that output while the cause is active. Empty = no action. STOP = output de-energised / burner request removed. RUN = energised. OPEN / CLOSE = resulting valve position. ALARM = HMI alarm and fault lamp. Column 'Source' shows where each line comes from: Client = client document; PH proposal = proposed by PH, to be approved by the client.
Requested from client	On sheet 'C&E Matrix', fill the two yellow columns for every row: 'Client response' (Approved / Approved with comment / Rejected) and 'Client comment'. Rows marked 'PH proposal' need particular attention. All other cells are protected against accidental edits (no password). Use the filter on the header row, e.g. Source = PH proposal. Parameter values are on sheet 'Parameters' and are adjustable from the HMI.

Revision	Date	Description	By / Approved
A	26-10-2026	First issue for client review	PH / Eng. Ahmed Zain

CAUSE & EFFECT MATRIX

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No.	Tag	Cause	PLC address	Condition / setpoint	Type	EFFECTS								Reset	Source	Remarks	Client response	Client comment
						E1 BR01 heat request (TL)	E2 BR01 2nd stage (TR)	E3 BR02 start (L)	E4 BR02 high fire (T2)	E5 XV-401 Flare SV (NO)	E6 XV-402 Drain SV (NO)	E7 XV-403 Suction SV (NC)	E8 HMI alarm / fault lamp					
						Q0.0 / R1	Q0.1 / R2	Q0.5 / R6	Q0.6 / R7	Q0.2 / R3	Q0.3 / R4	Q0.4 / R5	HMI, H3 / H4					
EMERGENCY SHUTDOWN																		
C01	S0	E-STOP pressed	I1.0	NC1 contact open	Trip	STOP	STOP	STOP	STOP	OPEN	OPEN	CLOSE	ALARM	Release S0, then S2	PH proposal	Burners: also cut by hardwired S0 NC2/NC3 (client e-mail: ESD impact on burner ON/OFF). Scrubber valves to safe position on ESD = PH proposal.		
C02	—	PLC STOP / 24 VDC loss	—	All outputs de-energised	Inherent	STOP	STOP	STOP	STOP	OPEN	OPEN	CLOSE		Automatic on restore + S2	PH proposal	Result of valve types (NO / NO / NC) and relay outputs — no program needed.		
GAS SCRUBBER																		
C03	PSH-101	Scrubber high pressure	I0.0	Contact open (fail-safe)	Trip					OPEN	OPEN	CLOSE	ALARM	Manual (S2) after cleared	Client	Client e-mail: scrubber alarms/trips. Burners NOT tripped (confirmed by Eng. Ahmed Zain).		
C04	LSH-102	Scrubber high level	I0.1	Contact open (fail-safe)	Alarm								ALARM	Automatic when cleared	Client	Pre-alarm before LSHH-103.		
C05	LSHH-103	Scrubber high-high level	I0.2	Contact open (fail-safe)	Trip					OPEN	OPEN	CLOSE	ALARM	Manual (S2) after cleared	Client	Burners NOT tripped (confirmed by Eng. Ahmed Zain).		
C06	LSL (scrubber)	Scrubber low level	I12.5 (reserved)	—	Not installed									—	Client	Not installed — excluded from Rev. A. Input reserved for future use.		
BURNER BR01 — tank TK01																		
C07a	LSL-TK01	TK01 level below low level switch (not ready)	I1.2	LSL not reached (contact open)	Permissive	STOP	STOP							Auto (AUTO mode)	Client	Client sheet: burner may start only when level has reached LSL. Contact closes with liquid present, so a broken wire also blocks the burner.		
C07b	TE-501 / TE-502	TK01 oil temperature at setpoint	IW96 / IW98	T >= SP (70 °C)	Control	STOP	STOP							Auto when T <= SP − HYST (AUTO mode)	Client	Client cases 2/3/6. Higher of inlet/outlet is used (PH proposal). Restart after hysteresis avoids short-cycling: each start costs ~50 s of purge.		
C07c	TE-501 / TE-502	TK01 oil temperature below 2nd-stage threshold	IW96 / IW98	T < SP − STG (60 °C)	Control		RUN							Auto when T >= SP − STG	PH proposal	Requests 2nd stage / high fire for fast heat-up. Between 60 and 70 °C: 1st stage only.		
C07d	TE-501 / TE-502	Thermocouple fault (wire break / out of range)	IW96 / IW98	SM 1231 diagnostic or value out of range	Trip	STOP	STOP						ALARM	Manual (S2) after fault cleared	PH proposal	Without a valid temperature the burner cannot be controlled.		
C07e	Rflt	BR01 burner lockout (flame failure, ignition failure, air pressure)	I0.5	Lockout relay energised	Alarm	STOP	STOP						ALARM	At burner control box, then S2	PH proposal	Burner's own control box has already shut the gas valves. PLC removes the heat request so the burner does not restart by itself after a local reset.		
C07f	RR	BR01 failed to start	I0.3	Heat request ON and no RUN feedback within T_START (120 s)	Trip	STOP	STOP						ALARM	Manual (S2)	PH proposal	Normal start cycle is ~55 s (RS100 manual §6.8.1) / ~65 s (DLX 53 manual).		
C07g	S4	BR01 selector in MANUAL	I12.1	Selector open = MAN	Mode									—	Client	Client e-mail: Auto/Manual selection. MAN: operator starts with START and stops with STOP. All trips and the temperature setpoint (C07b) still apply; no automatic restart.		
C07h	S1	BR01 START push button (MAN mode)	I0.4	Pressed	Command	RUN								—	PH proposal	MAN: starts the burner. AUTO: re-enables automatic control after STOP. Ignored while any trip or permissive is active.		
C07i	S3	BR01 STOP push button	I0.6	Pressed	Command	STOP	STOP							—	PH proposal	Works in AUTO and MAN. In AUTO the burner stays off until START is pressed.		
C07j	—	BR01 running continuously for 24 h	—	Run timer >= 24 h	Control	STOP	STOP							Automatic (restarts on next demand)	PH proposal	RS100 is approved for intermittent operation and must stop at least once every 24 h (RS100 manual §5.10).		
BURNER BR02 — tank TK02																		
C08a	LSL-TK02	TK02 level below low level switch (not ready)	I1.3	LSL not reached (contact open)	Permissive			STOP	STOP					Auto (AUTO mode)	Client	Client sheet: burner may start only when level has reached LSL. Contact closes with liquid present, so a broken wire also blocks the burner.		
C08b	TE-503 / TE-504	TK02 oil temperature at setpoint	IW100 / IW102	T >= SP (70 °C)	Control			STOP	STOP					Auto when T <= SP − HYST (AUTO mode)	Client	Client cases 2/3/6. Higher of inlet/outlet is used (PH proposal). Restart after hysteresis avoids short-cycling: each start costs ~50 s of purge.		
C08c	TE-503 / TE-504	TK02 oil temperature below 2nd-stage threshold	IW100 / IW102	T < SP − STG (60 °C)	Control				RUN					Auto when T >= SP − STG	PH proposal	Requests 2nd stage / high fire for fast heat-up. Between 60 and 70 °C: 1st stage only.		
C08d	TE-503 / TE-504	Thermocouple fault (wire break / out of range)	IW100 / IW102	SM 1231 diagnostic or value out of range	Trip			STOP	STOP				ALARM	Manual (S2) after fault cleared	PH proposal	Without a valid temperature the burner cannot be controlled.		
C08e	Rflt2	BR02 burner lockout (flame failure, ignition failure, air pressure)	I1.5	Lockout relay energised	Alarm			STOP	STOP				ALARM	At burner control box, then S2	PH proposal	Burner's own control box has already shut the gas valves. PLC removes the heat request so the burner does not restart by itself after a local reset.		
C08f	RR2	BR02 failed to start	I1.4	Heat request ON and no RUN feedback within T_START (120 s)	Trip			STOP	STOP				ALARM	Manual (S2)	PH proposal	Normal start cycle is ~55 s (RS100 manual §6.8.1) / ~65 s (DLX 53 manual).		
C08g	S5	BR02 selector in MANUAL	I12.2	Selector open = MAN	Mode									—	Client	Client e-mail: Auto/Manual selection. MAN: operator starts with START and stops with STOP. All trips and the temperature setpoint (C08b) still apply; no automatic restart.		
C08h	S6	BR02 START push button (MAN mode)	I12.3	Pressed	Command			RUN						—	PH proposal	MAN: starts the burner. AUTO: re-enables automatic control after STOP. Ignored while any trip or permissive is active.		
C08i	S7	BR02 STOP push button	I12.4	Pressed	Command			STOP	STOP					—	PH proposal	Works in AUTO and MAN. In AUTO the burner stays off until START is pressed.		
C08j	Rfuel2	BR02 in diesel mode	I12.0	Relay energised	Indication									—	Client	Read only: fuel is selected on the burner itself. Shown on HMI.		
GENERAL																		
C09	S2	FAULT RESET push button	I1.1	Pressed	Command									—	PH proposal	Resets latched trips only when their cause has cleared.		

ADJUSTABLE PARAMETERS (HMI)

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Parameter	Description	Default	Unit	Range	Source
SP	Tank temperature setpoint — burner stops at T >= SP	**70**	°C	40 – 90	Client sheet (adjustable from SCADA/HMI)
HYST	Restart hysteresis — burner restarts at T <= SP − HYST	**5**	°C	3 – 10	PH proposal: > type-K accuracy (±2.2 °C), limits short-cycling
STG	2nd stage / high fire while T < SP − STG	**10**	°C	5 – 30	PH proposal
T_START	Max. time from heat request to RUN feedback	**120**	s	90 – 300	PH proposal: ~2 × burner start cycle
T_MAXRUN	Forced stop of BR01 after continuous run	**24**	h	fixed	RS100 manual §5.10

Blue values are defaults; they can be changed from the HMI by an authorised user.

I/O LIST (from drawing PH-LCP-01 V1, sheets 10–13)

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Address	Tag	Description	Terminal	Signal	Sheet
I0.0	PSH-101	Scrubber high pressure	XI:1	DI 24V	10
I0.1	LSH-102	Scrubber high level	XI:3	DI 24V	10
I0.2	LSHH-103	Scrubber high-high level	XI:5	DI 24V	10
I0.3	RR	BR01 run feedback	XI:7	DI 24V	10
I0.4	S1	BR01 START push button	XI:9	DI 24V	10
I0.5	Rflt	BR01 fault (lockout)	XI:11	DI 24V	10
I0.6	S3	BR01 STOP push button	XI:13	DI 24V	10
I0.7	—	Spare	XI:15	DI 24V	10
I1.0	S0	E-STOP status (NC1)	XI:17	DI 24V	10
I1.1	S2	FAULT RESET push button	XI:19	DI 24V	10
I1.2	LSL-TK01	TK01 low level	XI:21	DI 24V	10
I1.3	LSL-TK02	TK02 low level	XI:23	DI 24V	10
I1.4	RR2	BR02 run feedback	XI:25	DI 24V	10
I1.5	Rflt2	BR02 fault (lockout)	XI:27	DI 24V	10
I12.0	Rfuel2	BR02 diesel mode	XI2:1	DI 24V	11
I12.1	S4	BR01 AUTO/MAN selector (closed = AUTO)	XI2:3	DI 24V	11
I12.2	S5	BR02 AUTO/MAN selector (closed = AUTO)	XI2:5	DI 24V	11
I12.3	S6	BR02 START push button	XI2:7	DI 24V	11
I12.4	S7	BR02 STOP push button	XI2:9	DI 24V	11
I12.5	—	Reserved: scrubber LSL (not installed)	XI2:11	DI 24V	11
I12.6–7	—	Spare	XI2:13–16	DI 24V	11
Q0.0	R1	BR01 heat request (TL) + H1 1st stage lamp	XQ:1/2	DQ 24V	12
Q0.1	R2	BR01 2nd stage (TR) + H2 2nd stage lamp	XQ:3/4	DQ 24V	12
Q0.2	R3	XV-401 Flare SV (NO)	XQ:5/6	DQ 24V	12
Q0.3	R4	XV-402 Drain SV (NO)	XQ:7/8	DQ 24V	12
Q0.4	R5	XV-403 Suction SV (NC)	XQ:9/10	DQ 24V	12
Q0.5	R6	BR02 start (L)	X6:2	DQ 24V	12 / 8
Q0.6	R7	BR02 high fire (T2)	X6:4	DQ 24V	12 / 8
Q0.7	R8	BR02 1st stage lamp H5	—	DQ 24V	12
Q1.0	R9	BR02 2nd stage lamp H6	—	DQ 24V	12
Q1.1	R10	Spare	XQ:19/20	DQ 24V	12
IW96	TE-501	TK01 inlet oil temperature	X5:1/2	TC type K	13
IW98	TE-502	TK01 outlet oil temperature	X5:3/4	TC type K	13
IW100	TE-503	TK02 inlet oil temperature	X5:5/6	TC type K	13
IW102	TE-504	TK02 outlet oil temperature	X5:7/8	TC type K	13
IW104–110	—	Spare thermocouple channels	X5:9–16	TC	13